

GANESH ARIVOLI

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SUMMARY

Graduate student in Mechanical Engineering with experience in software development, simulation and modeling of dynamic systems. Experienced in developing high-performance, physics-based simulation frameworks and algorithms for robotic and autonomous systems. Skilled in C++, Python and CUDA, with strong foundations in multi-body dynamics, control, and high-performance computing. Actively contributing to [Project Chrono](#) with a focus on GPU-accelerated multibody dynamics for robotics simulation.

EDUCATION

University of Wisconsin-Madison <i>Master of Science, Mechanical Engineering (Research Track)</i>	Aug 2024 – Dec 2026 <i>Current GPA: 4.0/4.0</i>
National Institute of Technology, Tiruchirappalli <i>Bachelor of Technology, Mechanical Engineering</i>	Jul 2017 – May 2021

RESEARCH EXPERIENCE

Graduate Research Assistant <i>Simulation-Based Engineering Lab, UW-Madison</i>	Aug 2024 – Present
- Contributing to the development of Project Chrono, an open-source C++ multiphysics simulation engine, implementing GPU-parallelized modules to accelerate multibody dynamics and terrain modeling for robotic systems. - Investigated the feasibility of using GPU-based linear solvers in multibody dynamics simulations. Benchmarked them against CPU solvers to quantify performance gains and numerical accuracy. Paper here (presented at <i>ASME IDETC-CIE 2025</i>) - Developed a CNN-based reduced-order model for autonomous terrain leveling with pipelines for training and deployment on HPC clusters achieving 48% improvement in terrain leveling performance. Paper here (submitted to <i>Advanced Robotics Research 2025</i>) - Collaborated on the development and testing of Chrono::CRM, a scalable GPU-accelerated terrain model to support autonomous vehicle planning and control simulations. Paper here (submitted to <i>Journal of Terramechanics 2025</i>).	
Vehicle Dynamics Engineer (R&D) <i>Bajaj Auto, India</i>	Sep 2021 – Dec 2023
- Led the end-to-end development of suspension and steering systems for Bajaj Auto-Triumph motorcycles from initial concept to final mass production, working with cross-functional teams and contributing to 100,000+ units sold across 58+ countries (product site). - Improved ride and handling performance by 15% through multibody dynamics simulations (MSC Adams), optimizing suspension geometry, spring rates, and damping coefficients integrating physics-based simulation models into the product development cycle.	

PROJECTS

GPU Accelerated Path Planning for Autonomous Robots <i>CS 759, UW-Madison</i>	Jan 2025 – Apr 2025
- Implemented parallelized RRT algorithm using CUDA (for GPU) and OpenMP (for multi-core CPU) targeting computationally intensive nearest-neighbor searches and collision detection operations for planning in obstacle-dense environments. - Profiled performance bottlenecks using NVIDIA Nsight Systems, identifying and improving runtime efficiency for the GPU-accelerated modules achieving 5X speedup over the CPU baseline. Github repo here.	
Robotic Manipulator Simulation and Control Toolkit <i>ECE 441, UW-Madison</i>	Jan 2025 – Apr 2025
- Developed a MATLAB-based simulation toolkit for kinematics, dynamics, and control of serial robotic manipulators using Euler-Lagrange equations, supporting multi-DOF systems with numerical integration and real-time visualization of robot motion. - Implemented motion planning and trajectory generation modules to compute smooth, collision-free joint-space paths under kinematic and dynamic constraints, incorporating controller bandwidth and actuation limits for execution feasibility.	
SAE BAJA Racing Team <i>NITT, India</i>	May 2018 – May 2021
- Spearheaded powertrain and data acquisition teams for an All-Terrain Vehicle, earning top-5 finish in BAJA SAE India 2020 and 2021. - Developed a MATLAB-based CVT model for tuning shift logic, improving efficiency by 10% and reducing tuning time by 40%.	

TECHNICAL SKILLS

Programming: C/C++, Python, MATLAB, CUDA, OpenMP, Bash, PyTorch
Tools & Platforms: Git/GitHub, Linux, NVIDIA Nsight, Docker, ROS, Make/CMake
Simulation: Simulink, Project Chrono, Mujoco, NVIDIA Isaac Sim, PyBullet, MSC Adams